II Tutorial

OLS estimation, inference and omitted variable bias

Exercise 1. Unbiasedness and consistency

Consider a cross-sectional model with one regressor, where i indexes individuals:

$$y_i = \beta_0 + \beta_1 x_i + u_i, \qquad i = 1, \dots, n.$$

Assume i.i.d. observations, finite fourth moments of x_i and u_i, and $\mathrm{Var}(x_i) > 0$. For finite-sample statements, consider samples with $\frac{1}{n}\sum_i (x_i - \bar{x})^2 > 0$ and assume the required expectations exist. Let $X = (x_1, \dots, x_n)$ denote the full sample of regressors.

To do.

(a) Show that the OLS slope estimator $\widehat{\beta}_1$ is unbiased under $E[u_i \mid X] = 0$. Derive its expression and show both $E[\widehat{\beta}_1 \mid X] = \beta_1$ and $E[\widehat{\beta}_1] = \beta_1$.

(b) Show that $\widehat{\beta}_1$ is consistent under the weaker assumptions $E[u_i] = 0$ and $\mathrm{Cov}(x_i, u_i) = 0$, instead of zero conditional mean. Derive its probability limit.

(c) Comment on the difference between unbiasedness and consistency.

Exercise 2. Simple and multiple regression: energy, market and oil

Notebook: Simple_Multiple_Regression.ipynb

Analyse monthly US energy-sector excess returns for 2011–2025 (180 months). Compare the market-only regression with the model including Brent spot-price changes. Interpret alpha and both slopes, including units and the meaning of holding the other regressor constant. Compare uncertainty and fit.

Relevant concepts

- Simple and multiple OLS; total and excess returns; intercept and slopes.
- Standard errors; hypotheses; t-statistics; p-values; confidence intervals; statistical versus economic significance.
- Partial associations; fitted values and residuals; R^2 and adjusted R^2; omitted variable bias; robust standard errors.

To do. Using the multiple-regression estimates, calculate fitted energy excess returns for Market = 1%, Brent = 10% and for Market = −1%, Brent = −10%. Decompose their difference (first minus second) into the Market and Brent contributions.

Exercise 3. Omitted variable bias

Suppose the population model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u, \qquad E[u \mid x_1, x_2] = 0.$$

Assume finite second moments and $\mathrm{Var}(x_1) > 0$. When x_2 is omitted, the population simple-regression slope is $\widetilde{\beta}_1 = \mathrm{Cov}(x_1, y)/\mathrm{Var}(x_1)$.

To do.

(a) Derive $\widetilde{\beta}_1$ in terms of β_1, β_2 and the covariance between the regressors.

(b) State when the omitted variable bias $\widetilde{\beta}_1 - \beta_1$ vanishes. Determine its sign when $\beta_2 > 0$ and $\mathrm{Cov}(x_1, x_2) > 0$.

Exercise 4. Reading a published regression: sports and stock returns

Edmans, García and Norli (2007) study national stock-market returns after international soccer matches. Their **sentiment hypothesis** is that losses worsen investors' mood and lower returns, with a weaker reaction to wins. Use their original regression on daily observations, including days without matches:

$$a_{it} = \beta_0 + \beta_W W_{it} + \beta_L L_{it} + v_{it}.$$

$W_{it} = 1$ on the first trading day after a win and 0 otherwise; L_{it} is defined analogously for a loss. The reference group has $W_{it} = L_{it} = 0$: it includes days without matches and any other observations not coded as a win or loss. Each slope compares its outcome with this group.

Dependent variable. a_{it} is the **abnormal daily log return** of the national stock-market index, in percent. *Abnormal* means that usual market and calendar effects have already been removed; this adjustment is already performed. These returns are not standardised. Coefficients measure differences in percentage points; one basis point equals 0.01 percentage points.

Term	Coefficient (pp)	Reported t	Approx. p-value
Intercept (β_0)	$\widehat{\beta}_0$	—	—
Win (β_W)	0.016	0.27	0.787
Loss (β_L)	−0.212	−3.27	0.0011

Original estimates: Table II, Panel A (abnormal raw returns), "All games", p. 1978. Sample: 182,919 country-day observations across 39 countries, including 638 wins and 524 losses. The intercept is left symbolic; the published table gives numerical estimates only for the win and loss indicators. Standard errors allow heteroskedasticity and contemporaneous cross-country correlation. P-values are added using two-sided Normal approximations from the published, rounded t-statistics.

Relevant concepts

- Abnormal returns; intercept and reference group; indicator coefficients; percentage points and basis points.
- Hypotheses; t-statistics; p-values; statistical versus economic significance.

To do. Explain the intercept's meaning and interpret the win and loss coefficients relative to the reference group. Express the fitted abnormal return in each case using $\widehat{\beta}_0$. Use $H_0 : \beta_j = 0$ versus $H_1 : \beta_j \neq 0$ ($j = W, L$) at 5%. Comment on the sentiment hypothesis, statistical and economic significance, and whether the results establish causality.

Source: Edmans, A., García, D. and Norli, Ø. (2007), *Sports Sentiment and Stock Returns*, *The Journal of Finance*, 62(4), 1967–1998. doi:10.1111/j.1540-6261.2007.01262.x.

Solutions

Exercise 1. Unbiasedness and consistency

(a) Unbiasedness

Start from the OLS slope:

$$\widehat{\beta}_1 = \frac{\frac{1}{n}\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n}\sum_i (x_i - \bar{x})^2}.$$

Using $y_i = \beta_0 + \beta_1 x_i + u_i$ and $\bar{y} = \beta_0 + \beta_1 \bar{x} + \bar{u}$, substitute $y_i - \bar{y} = \beta_1(x_i - \bar{x}) + (u_i - \bar{u})$:

$$\begin{aligned}
\widehat{\beta}_1 &= \frac{\frac{1}{n}\sum_i (x_i - \bar{x})[\beta_1(x_i - \bar{x}) + (u_i - \bar{u})]}{\frac{1}{n}\sum_i (x_i - \bar{x})^2} \\
&= \frac{\beta_1 \frac{1}{n}\sum_i (x_i - \bar{x})^2 + \frac{1}{n}\sum_i (x_i - \bar{x})(u_i - \bar{u})}{\frac{1}{n}\sum_i (x_i - \bar{x})^2} \\
&= \beta_1 + \frac{\frac{1}{n}\sum_i (x_i - \bar{x})(u_i - \bar{u})}{\frac{1}{n}\sum_i (x_i - \bar{x})^2}.
\end{aligned}$$

Finally, $\sum_i (x_i - \bar{x}) = 0$ eliminates the term involving $\bar{u}$:

$$\boxed{\widehat{\beta}_1 = \beta_1 + \frac{\frac{1}{n}\sum_i (x_i - \bar{x})u_i}{\frac{1}{n}\sum_i (x_i - \bar{x})^2}.}$$

Assume $E[u_i \mid X] = 0$. Conditioning on X fixes the regressors, so

$$E[\widehat{\beta}_1 \mid X] = \beta_1 + \frac{\frac{1}{n}\sum_i (x_i - \bar{x})E[u_i \mid X]}{\frac{1}{n}\sum_i (x_i - \bar{x})^2} = \beta_1.$$

Iterated expectations give $E[\widehat{\beta}_1] = \beta_1$. Homoskedasticity is not required; replacing an expectation of a ratio with a ratio of expectations would generally be incorrect.

(b) Consistency

By the law of large numbers and Slutsky's theorem,

$$\widehat{\beta}_1 = \beta_1 + \frac{\frac{1}{n}\sum_i x_i u_i - \bar{x}\bar{u}}{\frac{1}{n}\sum_i x_i^2 - \bar{x}^2} \xrightarrow{p} \beta_1 + \frac{\mathrm{Cov}(x_i, u_i)}{\mathrm{Var}(x_i)} = \beta_1.$$

Thus $\widehat{\beta}_1 \xrightarrow{p} \beta_1$. Zero covariance is sufficient here for consistency, but does not generally imply zero conditional mean or finite-sample unbiasedness.

(c) Unbiasedness versus consistency

Unbiasedness concerns the estimator's mean at a fixed n; consistency means $P(|\widehat{\beta}_1 - \beta_1| > \varepsilon) \to 0$ for every $\varepsilon > 0$ as $n \to \infty$. The conditions used here for consistency, $E[u_i] = 0$ and $\mathrm{Cov}(x_i, u_i) = 0$, are weaker than the zero conditional mean assumption used for unbiasedness. An estimator can therefore be consistent while biased in finite samples.

Exercise 3. Omitted variable bias

(a) Population derivation

Exogeneity gives $E[u] = 0$ and $E[x_1 u] = E\{x_1 E[u \mid x_1, x_2]\} = 0$, hence $\mathrm{Cov}(x_1, u) = 0$. Substitute the population model and expand the covariance:

$$\begin{aligned}\widetilde{\beta}_1 &= \frac{\mathrm{Cov}(x_1, y)}{\mathrm{Var}(x_1)} = \frac{E[(x_1 - E[x_1])(y - E[y])]}{E[(x_1 - E[x_1])^2]} \\ &= \frac{\mathrm{Cov}(x_1, \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u)}{\mathrm{Var}(x_1)} \\ &= \frac{\beta_1 \,\mathrm{Var}(x_1) + \beta_2 \,\mathrm{Cov}(x_1, x_2) + \mathrm{Cov}(x_1, u)}{\mathrm{Var}(x_1)} \\ &= \beta_1 + \beta_2 \frac{\mathrm{Cov}(x_1, x_2)}{\mathrm{Var}(x_1)}.\end{aligned}$$

Therefore,

$$\boxed{\widetilde{\beta}_1 - \beta_1 = \beta_2 \frac{\mathrm{Cov}(x_1, x_2)}{\mathrm{Var}(x_1)}.}$$

Under i.i.d. sampling, the short-regression OLS slope converges to $\widetilde{\beta}_1$. This population coefficient gap is its asymptotic bias relative to β_1; it is not a formula for finite-sample expected bias.

(b) Conditions and direction

The bias vanishes if $\beta_2 = 0$ or $\mathrm{Cov}(x_1, x_2) = 0$. With both positive, omitting x_2 raises the population slope on x_1.

Exercise 4. Reading a published regression: sports and stock returns

Coefficient interpretation and units

The estimated intercept $\widehat{\beta}_0$ is the fitted abnormal return for the reference group ($W_{it} = L_{it} = 0$), which includes days without matches but is not restricted to those days. Substituting the indicator values gives

$$\begin{aligned}\text{Reference } (0,0): &\quad \widehat{a}_{it} = \widehat{\beta}_0 + 0.016(0) - 0.212(0) = \widehat{\beta}_0, \\ \text{Win } (1,0): &\quad \widehat{a}_{it} = \widehat{\beta}_0 + 0.016(1) - 0.212(0) = \widehat{\beta}_0 + 0.016, \\ \text{Loss } (0,1): &\quad \widehat{a}_{it} = \widehat{\beta}_0 + 0.016(0) - 0.212(1) = \widehat{\beta}_0 - 0.212.\end{aligned}$$

Thus wins are associated with fitted abnormal returns **0.016 percentage points higher** (1.6 basis points), and losses with returns **0.212 percentage points lower** (21.2 basis points), relative to the reference group. These differences do not require setting $\widehat{\beta}_0$ to zero.

P-values and statistical significance

Test $H_0 : \beta_j = 0$ against $H_1 : \beta_j \neq 0$ for $j = W, L$. Under the null and test assumptions, the p-value is the probability of a statistic at least as extreme as observed, not the probability that H_0 is true. At 5%, **wins:** $p \approx 0.787$, do not reject the zero-coefficient null; this does not prove that the population win coefficient is zero. **Losses:** $p \approx 0.0011$, reject, with evidence of lower abnormal returns than in the reference group.

Economic interpretation and limitations

The negative loss coefficient and small, insignificant win coefficient are consistent with the stated sentiment hypothesis. The 21.2-basis-point loss difference describes economic magnitude; the p-value measures statistical evidence against zero. It does not mean that every loss produces a market decline of that size. The table alone does not establish that mood causes the return difference: economic consequences and other confounding factors remain possible explanations.